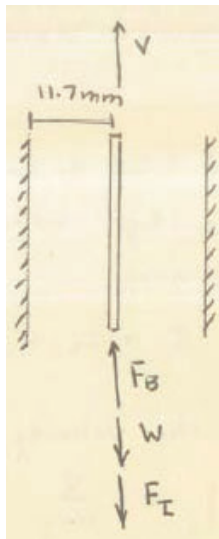
Problem 2.21

(Difficulty: 2)

2.21 A vertical gap 25 mm wide of infinite extent contains oil of specific gravity 0.95 and viscosity 2.4 Pa · s. A metal plate 1.5 m × 1.5 m × 1.6 mm weighting 45 N is to be lifted through the gap at a constant speed of 0.06 m/s.Estimate the force required.



Given: Plate size:1.5 m × 1.5 m × 1.6 mm .Width of gap: 25 mm. Specific gravity:0.95.

Viscosity: 2.4 Pa · s. Weight: 45 N. Speed: 0.06 m/s.

Find: The force required F_T.

Assumption: Viscous flow. Neglecting all end effects. Linear velocity profile in the gap.

Solution: Make a force balance on the plate. Use Newton's law of viscosity to relate the viscous force on the plate to the viscosity and velocity.

We need to calculate all the individual forces. There are the force due to gravity (weight), the buoyancy force, and the drag force.

Buoyancy force:

$$F_B = \rho g V = SG \cdot \rho_{H_2o} g V = 0.95 \times 998.2 \times 9.81 \times 1.5 \times 1.5 \times 0.0016 = 33.5\ N$$

Drag force: The viscous shear stress is given by

$$\tau = \mu \frac{du}{dx} = \mu \frac{\Delta u}{\Delta x}$$

The force on both sides of the plate is

$$F_\tau = 2\,\mu \frac{\Delta u}{\Delta x} A = 2 \times 2.4\,Pa\ \ s \times \frac{0.06\,\frac{m}{s}}{0.0117\,m} \times (1.5\,m)^2 = 55.4\,N$$

The force required to maintain motion is F_T, by the force balance equation we have:

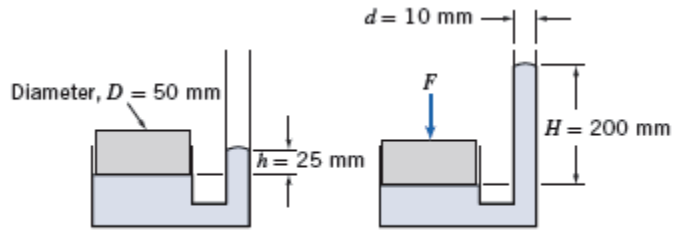
$$F_T + F_B = W + F_\tau$$

The total force is then

$$F_T = W + F_\tau - F_B = 45\,N + 55.4\,N - 33.5\,N = 66.9\,N$$

Problem 3.5 [Difficulty 2]

3.5 A piston is placed on a tank filled with mercury at 20 °C as shown below. A force is applied to the piston and the height of the mercury column rises. Determine the weight of the piston and the applied force.



P3.6

Given: Data on system before and after applied force

Find: Applied force

Solution:

Basic equation $\frac{dp}{dy} = -\rho \cdot g$ or, for constant ρ $p = p_{atm} - \rho \cdot g \cdot (y - y_0)$ with $p(y_0) = p_{atm}$

For initial state $p_1 = p_{atm} + \rho \cdot g \cdot h$ and $F_1 = p_1 \cdot A = \rho \cdot g \cdot h \cdot A$ (Gage; F_1 is hydrostatic upwards force)

For the initial FBD $\Sigma F_y = 0$ $F_1 - W = 0$ $W = F_1 = \rho \cdot g \cdot h \cdot A$

For final state $p_2 = p_{atm} + \rho \cdot g \cdot H$ and $F_2 = p_2 \cdot A = \rho \cdot g \cdot H \cdot A$ (Gage; F_2 is hydrostatic upwards force)

For the final FBD $\Sigma F_y = 0$ $F_2 - W - F = 0$ $F = F_2 - W = \rho \cdot g \cdot H \cdot A - \rho \cdot g \cdot h \cdot A = \rho \cdot g \cdot A \cdot (H - h)$

$$F = \rho_{H2O} \cdot SG \cdot g \cdot \frac{\pi \cdot D^2}{4} \cdot (H - h)$$

$$SG = 13.54$$

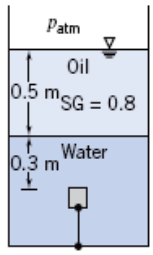
$$F = 1000 \cdot \frac{kg}{m^3} \times 13.54 \times 9.81 \cdot \frac{m}{s^2} \times \frac{\pi}{4} \times (0.05 \cdot m)^2 \times (0.2 - 0.025) \cdot m \times \frac{N \cdot s^2}{kg \cdot m}$$

$$F = 45.6N$$

Problem 3.6

[Difficulty: 2]

3.6 A 125-mL cube of solid oak is held submerged by a tether as shown. Calculate the actual force of the water on the bottom surface of the cube and the tension in the tether.



Given: Data on system

Find: Force on bottom of cube; tension in tether

Solution:

Basic equation $\frac{dp}{dy} = -\rho \cdot g$ or, for constant ρ $\Delta p = \rho \cdot g \cdot h$ where h is measured downwards

The absolute pressure at the interface is $p_{interface} = p_{atm} + SG_{oil} \cdot \rho \cdot g \cdot h_{oil}$

Then the pressure on the lower surface is $p_L = p_{interface} + \rho \cdot g \cdot h_L = p_{atm} + \rho \cdot g \cdot \left(SG_{oil} \cdot h_{oil} + h_L\right)$

For the cube $V = 125 \cdot mL$ $V = 1.25 \times 10^{-4} \cdot m^3$

Then the size of the cube is $d = V^{\frac{1}{3}}$ $d = 0.05\,m$ and the depth in water to the upper surface is $h_U = 0.3 \cdot m$

Hence $h_L = h_U + d$ $h_L = 0.35\,m$ where h_L is the depth in water to the lower surface

The force on the lower surface is $F_L = p_L \cdot A$ where $A = d^2$ $A = 0.0025\,m^2$

$$F_L = \left[p_{atm} + \rho \cdot g \cdot \left(SG_{oil} \cdot h_{oil} + h_L\right)\right] \cdot A$$

$$F_L = \left[101 \times 10^3 \cdot \frac{N}{m^2} + 1000 \cdot \frac{kg}{m^3} \times 9.81 \cdot \frac{m}{s^2} \times (0.8 \times 0.5 \cdot m + 0.35 \cdot m) \times \frac{N \cdot s^2}{kg \cdot m}\right] \times 0.0025 \cdot m^2$$

$F_L = 270.894\,N$ Note: Extra decimals needed for computing T later!

For the tension in the tether, an FBD gives $\Sigma F_y = 0$ $F_L - F_U - W - T = 0$ or $T = F_L - F_U - W$

where $F_U = \left[p_{atm} + \rho \cdot g \cdot \left(SG_{oil} \cdot h_{oil} + h_U\right)\right] \cdot A$

Note that we could instead compute $\Delta F = F_L - F_U = \rho \cdot g \cdot SG_{oil} \cdot \left(h_L - h_U\right) \cdot A$ and $T = \Delta F - W$

Using F_U

$$F_U = \left[101 \times 10^3 \cdot \frac{N}{m^2} + 1000 \cdot \frac{kg}{m^3} \times 9.81 \cdot \frac{m}{s^2} \times (0.8 \times 0.5 \cdot m + 0.3 \cdot m) \times \frac{N \cdot s^2}{kg \cdot m}\right] \times 0.0025 \cdot m^2$$

$F_U = 269.668\,N$ Note: Extra decimals needed for computing T later!

For the oak block $SG_{oak} = 0.77$ so $W = SG_{oak} \cdot \rho \cdot g \cdot V$

$$W = 0.77 \times 1000 \cdot \frac{kg}{m^3} \times 9.81 \cdot \frac{m}{s^2} \times 1.25 \times 10^{-4} \cdot m^3 \times \frac{N \cdot s^2}{kg \cdot m} \qquad W = 0.944\,N$$

$T = F_L - F_U - W$ $T = 0.282\,N$

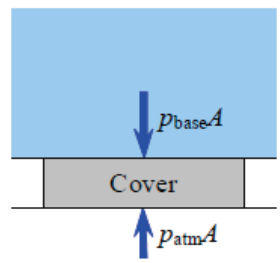
Problem 3.10 [Difficulty 2]

3.10 A water tank filled to a depth of 16 ft. has a 1 in. x 1 in. inspection cover at the base. The cover is held in place by a plastic bracket that can withstand a load of 9 lbf. Determine whether the bracket is strong enough under these conditions, and the water depth that would cause the bracket to break.

Given: Data on water tank and inspection cover

Find: If the support bracket is strong enough; at what water depth would it fail

Assumptions: Water is incompressible and static



Solution:

Basic equation $\frac{dp}{dy} = -\rho \cdot g$ or, for constant ρ $\Delta p = \rho \cdot g \cdot h$ where h is measured downwards

The absolute pressure at the base is $p_{base} = p_{atm} + \rho \cdot g \cdot h$ where $h = 16 \cdot ft$

The gage pressure at the base is $p_{base} = \rho \cdot g \cdot h$ This is the pressure to use as we have p_{atm} on the outside of the cover.

The force on the inspection cover is $F = p_{base} \cdot A$ where $A = 1 \cdot in \times 1 \cdot in$ $A = 1 \cdot in^2$

$$F = \rho \cdot g \cdot h \cdot A$$

$$F = 1.94 \cdot \frac{slug}{ft^3} \times 32.2 \cdot \frac{ft}{s^2} \times 16 \cdot ft \times 1 \cdot in^2 \times \left(\frac{ft}{12 \cdot in}\right)^2 \times \frac{lbf \cdot s^2}{slug \cdot ft}$$

$F = 6.94 \cdot lbf$ The bracket is strong enough (it can take 9 lbf).

To find the maximum depth we start with $F = 9.00 \cdot lbf$

$$h = \frac{F}{\rho \cdot g \cdot A}$$

$$h = 9 \cdot lbf \times \frac{1}{1.94} \cdot \frac{ft^3}{slug} \times \frac{1}{32.2} \cdot \frac{s^2}{ft} \times \frac{1}{in^2} \times \left(\frac{12 \cdot in}{ft}\right)^2 \times \frac{slug \cdot ft}{lbf \cdot s^2}$$

$$h = 20.7 \cdot ft$$

Problem 3.20 [Difficulty 2]

3.20 The elevation of Tucson, AZ, is about 500 m, and Mt. Lemmon is about 2000 m higher. Assuming standard atmospheric conditions in Tucson, determine the pressure at the top of Mt. Lemmon assuming (a) an incompressible atmosphere and (b) an atmosphere in which the temperature varies linearly with altitude. Compare your result to the value for a standard atmosphere.

Given: Elevations for two locations, atmospheric conditions.

Find: Pressure variation with altitiude under two assumptions.

Solution: Apply the relation between pressure and elevation.

The governing relation for the variation of pressure with elevation is Eq. 3.6

$$\frac{dp}{dz} = -\rho g$$

For standard atmospheric conditions in Tucson, from Table A.3, the pressure and density relative to sea level conditions are 0.9421 and 0.9529, respectively. Based on sea level conditions of 101.3 kPa and 1.225 kg/m3, the pressure and density at the elevation of Tucson are 95.4 kPa and 1.167 kg/m^3, respectively.

(a) For an incompressible atmosphere, the density is constant. Integrating Eq. 3.6 yields

$$\int_{p_1}^{p_2} dp = -\int_{z_1}^{z_2} \rho g \, dz$$

Or

$$p_2 - p_1 = -\rho g \left(z_2 - z_1 \right)$$

$$p_2 = p_1 - \rho g \left(z_2 - z_1 \right)$$

Using the value of ρ as the value in Tucson, the pressure at the top of Mt Lemmon is

$$p_2 = 95.4 kPa - 1.167 \frac{kg}{m^3} \cdot 9.81 \frac{m}{s^2} \left(2500 m - 500 m_1 \right) \cdot \frac{kPa}{1000 Pa}$$

$$= 72.6 \, kPa$$

(b) For an atmosphere in which the temperature varies lineatly with altitude, the pressure is given by equation 3.9:

$$p_2 = p_1 \left(\frac{T_2}{T_1} \right)^{g/mR}$$

Where m is the rate of change with temperature with altitude as given by the relation

$$T_2 = T_1 - m \left(z_2 - z_1 \right)$$

Using the Table A.3, the temperature at Tucson for the standard atmosphere is 284.9 K and that at the elevation of Mt Lemmon is 271.9K. The value of m is then"

$$m = -\frac{T_1 - T_2}{z_1 - z_2} = \frac{(284.9 - 271.9)K}{(500 - 2500)m} = 0.0065\frac{K}{m}$$

The value of R from Table A.6 is $286.9\frac{J}{kg\,K}$ The value of the exponent s then

$$\frac{g}{mR} = \frac{9.81\frac{m}{s^2}}{0.0065\frac{K}{m}\cdot 286.9\frac{J}{kg\,K}} = 5.255$$

The pressure at the top of Mt. Lemmon is then \

$$p_2 = p_1\left(\frac{T_2}{T_1}\right)^{g/mR} = 95.4kPa\left(\frac{271.9K}{284.9K}\right)^{5.255} = 74.7kPa$$

From Table A.3, the relative pressure at 2500 m is 0.7372. This corresponds to a pressure of 74.7 kPa. The assumption that the temperature varies linearly with altitude is a good assumption for lower elevations.

Problem 3.40

[Difficulty: 2]

3.40 A hydrometer is a specific gravity indicator, the value being indicated by the level at which the free surface intersects the stem when floating in a liquid. The 1.0 mark is the level when in distilled water. For the unit shown, the immersed volume in distilled water is 15 cm^3. The stem is 6 mm in diameter. Find the distance, h, from the 1.0 mark to the surface when the hydrometer is placed in a nitric acid solution of specific gravity 1.5.

Given: Hydrometer as shown, submerged in nitric acid. When submerged in water, h = 0 and the immersed volume is 15 cubic cm.

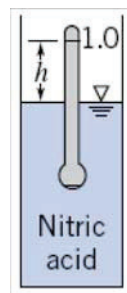
$SG = 1.5 \quad d = 6 \cdot mm$

Find: The distance h when immersed in nitric acid.

Solution: We will apply the hydrostatics equations to this system.

Governing Equations: $F_{buoy} = \rho \cdot g \cdot V_d$ (Buoyant force is equal to weight of displaced fluid)

Assumptions: (1) Static fluid
(2) Incompressible fluid



Taking a free body diagram of the hydrometer: $\Sigma F_z = 0 \quad -M \cdot g + F_{buoy} = 0$

Solving for the mass of the hydrometer: $M = \dfrac{F_{buoy}}{g} = \rho \cdot V_d$

When immersed in water: $M = \rho_w \cdot V_w$ When immersed in nitric acid: $M = \rho_n \cdot V_n$

Since the mass of the hydrometer is the same in both cases: $\rho_w \cdot V_w = \rho_n \cdot V_n$

When the hydrometer is in the nitric acid: $V_n = V_w - \dfrac{\pi}{4} \cdot d^2 \cdot h \quad \rho_n = SG \cdot \rho_w$

Therefore: $\rho_w \cdot V_w = SG \cdot \rho_w \cdot \left(V_w - \dfrac{\pi}{4} \cdot d^2 \cdot h\right)$ Solving for the height h:

$$V_w = SG \cdot \left(V_w - \frac{\pi}{4} \cdot d^2 \cdot h\right) \qquad V_w \cdot (1 - SG) = -SG \cdot \frac{\pi}{4} \cdot d^2 \cdot h$$

$$h = V_w \cdot \left(\frac{SG - 1}{SG}\right) \cdot \frac{4}{\pi \cdot d^2} \qquad h = 15 \cdot cm^3 \times \left(\frac{1.5 - 1}{1.5}\right) \times \frac{4}{\pi \times (6 \cdot mm)^2} \times \left(\frac{10 \cdot mm}{cm}\right)^3 \qquad h = 177 \cdot mm$$